

Assignment 01 – The Multi-Locator Challenge

1. Module Name

Module 1 – Automation with Selenium

2. Experiment Title

The Multi-Locator Challenge

3. Problem Statement

Navigate to a login page and automate the login process using different Selenium locator strategies. Use ID for the username, NAME for the password, and XPATH for the Login button. Verify that the resulting URL contains /inventory.html.

4. Objective

To understand and demonstrate Selenium WebDriver element identification using multiple locator strategies and validate a successful login.

5. Tools, Software and Concepts Used

- Python 3.x
- Selenium WebDriver
- Google Chrome
- Visual Studio Code
- By.ID
- By.NAME
- By.XPATH
- send_keys()
- click()
- assert statement
- driver.current_url

6. Source Code / Implementation Steps

1. Open the SauceDemo login page.
2. Locate the username field using ID and enter the username.
3. Locate the password field using NAME and enter the password.
4. Locate the Login button using XPATH and click it.
5. Verify that /inventory.html is present in the current URL.
6. Print the successful execution details and close the browser.

Source Code

```
from selenium import webdriver
```

```
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()

try:
    driver.get("https://www.saucedemo.com/")
    driver.maximize_window()

    username = driver.find_element(By.ID, "user-name")
    username.send_keys("standard_user")

    password = driver.find_element(By.NAME, "password")
    password.send_keys("secret_sauce")

    login_button = driver.find_element(By.XPATH, "//input[@id='login-button']")
    login_button.click()

    assert "/inventory.html" in driver.current_url

    print("Login successful.")
    print("Current URL:", driver.current_url)
    print("Assignment 01 completed successfully.")

    input("Press ENTER to close the browser...")

finally:
    driver.quit()
```

7. Output / Labelled Screenshots

- Screenshot 1 – SauceDemo Login Page
- Screenshot 2 – Inventory Page after successful login
- Screenshot 3 – VS Code terminal showing successful execution

Screenshots are stored in the Screenshots folder for this assignment.

8. Result

The login process was automated successfully using ID, NAME and XPATH locators. The resulting URL was verified to contain /inventory.html.

9. Observation

Different locator strategies can be used to identify web elements. ID provides direct identification, NAME can identify elements through the name attribute, and XPATH provides flexible attribute-based identification.

10. Conclusion

The experiment successfully demonstrated basic Selenium WebDriver automation and multiple locator strategies. The login workflow was automated and the result was verified using the current URL.